

IoT RELAY TO CONTROL APPLIANCES Through The Internet Using Smartphone

PRADIPTA GHOSH, CHALLA NAGA SAI RAVALI

Electrical appliances such as lights, fans, air-conditioners, TVs, and water pumps are

typically operated using manual switches. This system enables these appliances to be controlled over

the internet using a smartphone. It combines an ESP-01 Wi-Fi module, the Switchsys MQTT broker (free public version), and the IoT MQTT panel app. The Switchsys MQTT broker implements the MQTT publish/subscribe communication model, enabling communication between the ESP-01 module and the user interface.

The IoT MQTT panel mobile application allows users to send on/off commands and monitor device status in real time from any location with internet access. Commands sent through the app are routed via the MQTT broker to the ESP-01, which processes the subscribed topic and drives the relay accordingly. This enables remote control of connected appliances from virtually anywhere. Fig. 1 shows the author's prototype, while Table 1 lists the components required to build the system.

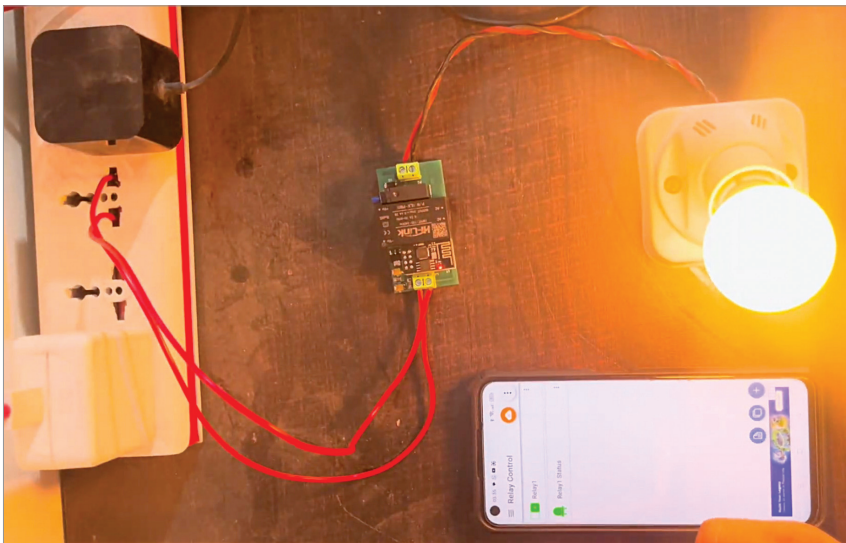


Fig. 1: Authors' prototype

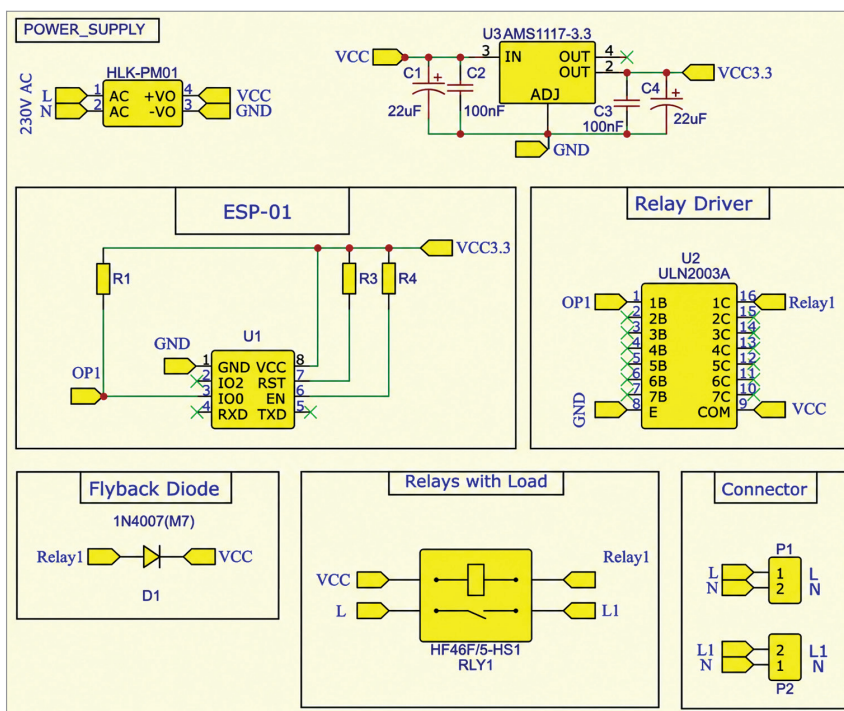


Fig. 2: Circuit diagram

TABLE 1
BILL OF MATERIALS

Components	Quantity
ESP-01 Wi-Fi module (ESP8266 SoC based)	1
8-pin ESP-01 connector	1
ESP-01 USB programmer adaptor	1
Relay driver IC ULN2003A (U2)	1
Relay HF46F/5-HS1	1
Diode 1N4007 (M7)	1
HCLK-PM01 230V AC to 5V DC converter	1
10kΩ resistor (R1-R3)	3
2-pin 5.08mm pitch plug-in screw terminal block connector	2
AMS1117-3.3 voltage regulator - U3 (5V DC to 3.3V DC)	1
100nF ceramic capacitor (C2, C3)	2
22μF tantalum capacitor (C1, C4)	2